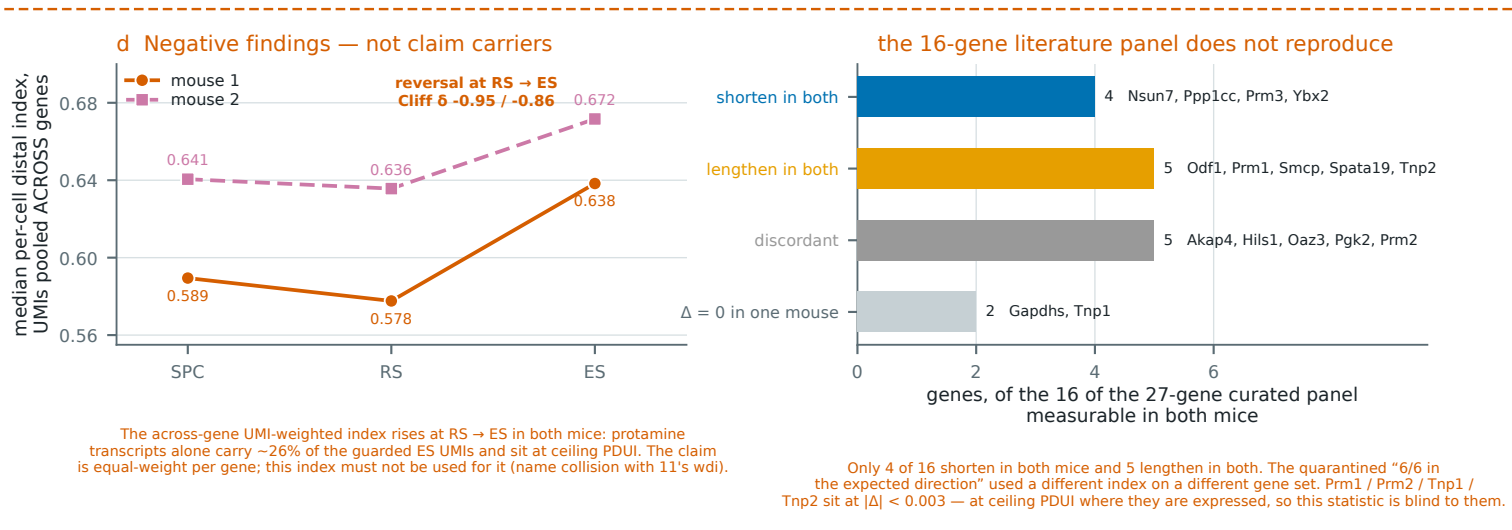
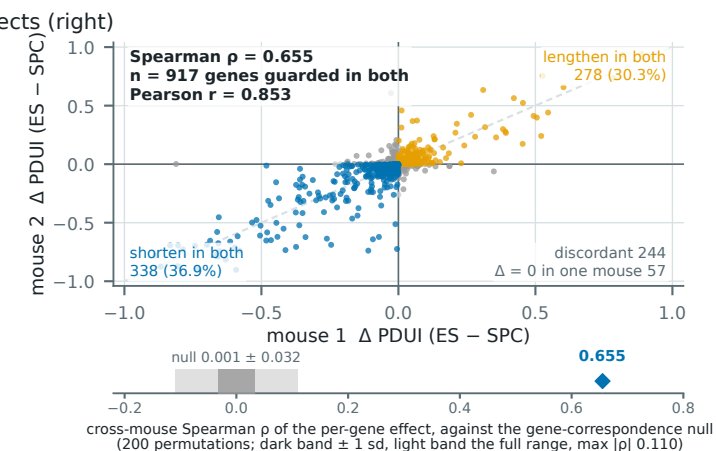
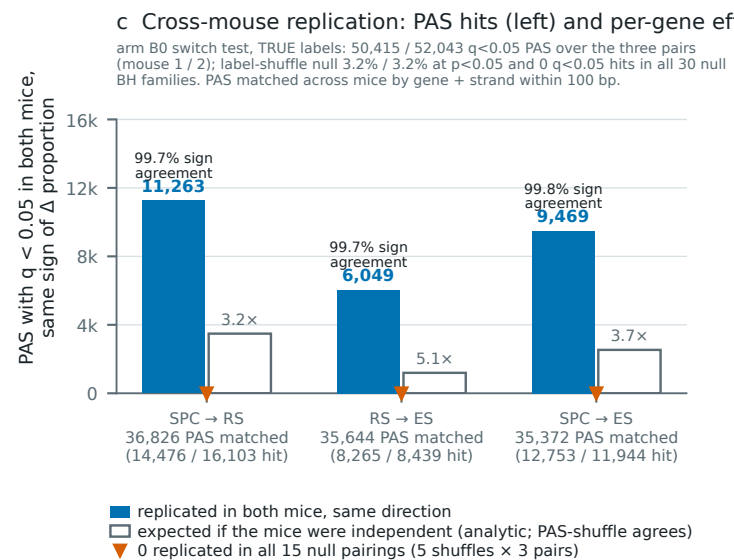
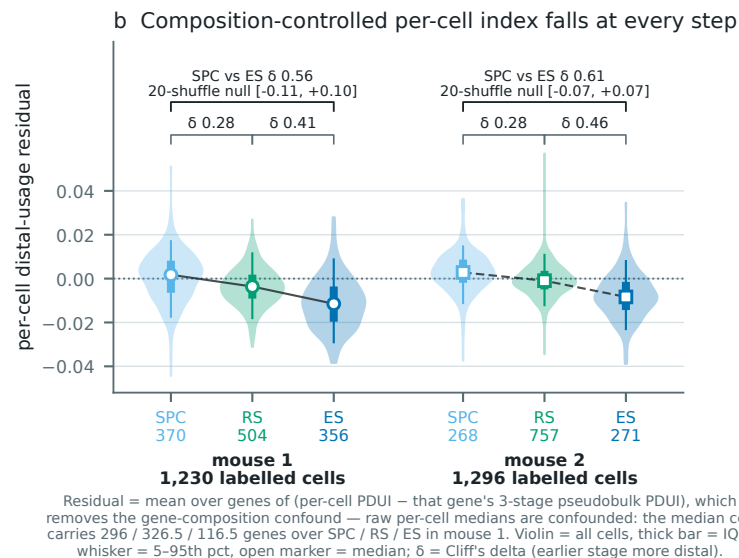
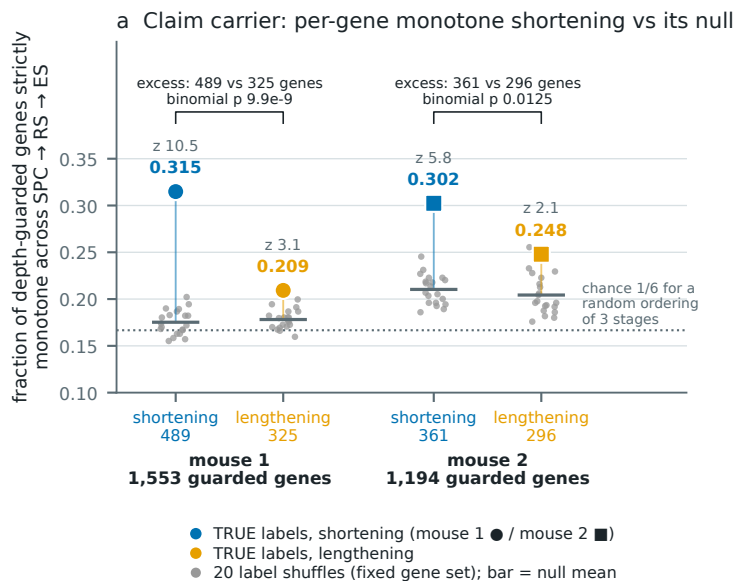


Per-gene 3'UTR usage shifts progressively along spermatocyte → round → elongating spermatid, in both mice

GSE104556 testis, both mice, PeakATail final caller (frozen tree 4efeb125); GEX marker-argmax stage labels; PDUI = distal / (proximal + distal) per gene, pseudobulk per stage, over genes with ≥ 50 UMI at the gene's proximal/distal PAS pair in every stage. 31.5% and 30.2% of guarded genes (489 / 1,553 and 361 / 1,194) shorten monotonically across the three stages, against label-shuffle nulls of 17.5% and 21.0% ($z = 10.5$ and 5.8 , 20 shuffles), exceeding monotone lengthening (489 vs 325 and 361 vs 296 genes); the composition-controlled per-cell distal-usage residual falls at every step (Cliff's δ SPC vs ES = 0.56 and 0.61, outside the whole 20-shuffle null range). This is a per-gene, EQUAL-WEIGHT statement (panel d).



Cautions, binding on any use of this figure. (1) Monotone LENGTHENING is also above its null ($z = 3.1$ / 2.1): the shuffle destroys ordered structure of either sign, so the direction-specific evidence is the EXCESS of shortening over lengthening — strong in mouse 1 (binomial p 9.9e-9), modest in mouse 2 (p 0.0125; 1 of 40 null draws reached the same excess in absolute value). (2) The per-gene MEDIANS are not monotone (RS marginally above SPC in both mice: $0.787 \rightarrow 0.794$ and $0.878 \rightarrow 0.886$); the ordering is carried by the monotone fraction, the paired per-gene tests and the per-cell residual, not by medians, and not by any UMI-weighted index (panel d). Wilcoxon p-values are quoted only next to their shuffle-null column and are therefore not on this figure. (3) SPG is excluded: SPG → SPC LENGTHENS under exactly these marker-argmax labels and rests on 64 / 68 cells (11). (4) Two mice of one study (GSE104556), one protocol — this controls animal-level and sampling noise, not study- or protocol-level artefacts. Stage labels are marker-panel argmax over GEX leiden clusters (98.8% agreement with independent PAS-cluster labels), not ground truth. (5) The ≥ 50 -UMI depth guard keeps 1,553 / 1,194 of ~10,400 genes with a PAS pair, biased toward highly expressed genes; the null uses the same guarded set. (6) The switch test's null is conservative, not exact (Fisher discreteness), so TRUE hit counts are not "N true switches"; only 5 label shuffles per mouse there (20 for the PDUI arm). Cross-mouse PAS matching is coordinate-based (100 bp, greedy 1:1), so only the matched subset can replicate. (7) PROVISIONAL: code is the frozen tree 4efeb125, and a v2 re-run after PRs #96 (IP-filter/strand) and #97 will redo every number here. Sources: manuscript/18_spermatogenesis_final.md (verdict FIXED, 2026-08-21) and results/stage3_spermatogenesis_final/ (REPORT_PROVISIONAL.md §2-4, summary/ and mouse{1,2}/summary/). Every plotted value: results/figures/manuscript/spermatogenesis_final*.tsv.